

Better skills data for smarter financing of education and training



Disclaimers

This work is issued under the responsibility of the Secretary-General of the OECD and does not necessarily reflect the official views of OECD Member countries.

This document, as well as any data and map included herein, are without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area.

Photo credits: © Andrey_Popov/Shutterstock.com.

© OECD 2026



Attribution 4.0 International (CC BY 4.0)

This work is made available under the Creative Commons Attribution 4.0 International licence. By using this work, you accept to be bound by the terms of this licence (<https://creativecommons.org/licenses/by/4.0/>).

Attribution – you must cite the work.

Translations – you must cite the original work, identify changes to the original and add the following text: *In the event of any discrepancy between the original work and the translation, only the text of the original work should be considered valid.*

Adaptations – you must cite the original work and add the following text: *This is an adaptation of an original work by the OECD. The opinions expressed and arguments employed in this adaptation should not be reported as representing the official views of the OECD or of its Member countries.*

Third-party material – the licence does not apply to third-party material in the work. If using such material, you are responsible for obtaining permission from the third party and for any claims of infringement.

You must not use the OECD logo, visual identity or cover image without express permission or suggest the OECD endorses your use of the work.

Any dispute arising under this licence shall be settled by arbitration in accordance with the Permanent Court of Arbitration (PCA) Arbitration Rules 2012. The seat of arbitration shall be Paris (France). The number of arbitrators shall be one.

Abstract

Robust and integrated skills data are essential for effective policymaking, particularly in improving the allocation of public resources across education and training. Yet many governments face significant information gaps. Fragmented or underdeveloped data systems constrain the ability to assess programme effectiveness, prioritise spending, and target investments towards high-impact interventions.

This paper examines how stronger skills data can support more evidence-based financing decisions. It introduces a framework grouping the main obstacles into four categories: institutional, governance and financing barriers; human-capital and analytical-capacity gaps; legal and regulatory constraints; and technical and interoperability challenges.

By analysing these barriers and highlighting promising country practices, the paper provides a basis for identifying priorities and guiding reforms to strengthen skills data systems and improve investment in skills.

Contact

Andrew BELL (✉ Andrew.BELL@oecd.org)

Ricardo ESPINOZA (✉ Ricardo.ESPINOZA@oecd.org)

Table of contents

Disclaimers	2
Abstract	3
Executive summary	6
1 Introduction	7
1.1. Why better data are essential	7
1.2. Purpose and scope	8
2 Current data gaps and their consequences	10
2.1. Fragmentation of data systems	10
2.2. Lack of longitudinal linkages	11
2.3. Coverage gaps across the skills system	12
2.4. Inconsistent definitions and quality issues	13
2.5. Timeliness and accessibility	13
2.6. Granularity: Equity and regional blind spots	14
3 Types of data sources for decision making, and what they can and cannot tell policymakers	15
4 What is at stake: Potential gains from closing the gaps	18
4.1. Sharper financial decision making: Reallocating to high-return programmes, avoiding low-yield spend, managing fiscal risk	18
4.2. Economic and social returns over the short and long term	19
5 Barriers to building integrated skills data systems	20
5.1. Institutional and governance barriers	20
5.2. Human-capital and analytical-capacity gaps	22
5.3. Legal and regulatory constraints	24
5.4. Technical and interoperability challenges	26

6 Conclusion	30
Annex A. Types of data sources for decision making	31
References	39

FIGURES

Figure 1. A framework for understanding barriers to integrated skills data systems	9
Figure 2. Analytical strengths of data sources for education and skills policymaking	17

TABLE

Table 1. Summary table: Data sources for decision making	16
--	----

Executive summary

Robust and integrated skills data are essential for effective policymaking, particularly in improving the allocation of public resources across education and training. Yet many governments continue to face significant information gaps that constrain their ability to assess programme effectiveness, prioritise spending and target investments towards interventions that yield the strongest outcomes.

This paper examines how stronger skills data can support more evidence-based and efficient financing decisions in education and training systems. It begins by identifying the main data gaps that currently hinder strategic decision making, including fragmentation of data systems across ministries and levels of government, limited longitudinal linkages between education and labour market records, uneven coverage across the skills system – particularly in early childhood and adult learning – and persistent issues with timeliness, accessibility, definitional consistency and data quality. These gaps leave policymakers without a consolidated view of how public resources translate into results, limiting their capacity to respond to evolving labour market needs.

The paper then reviews ten data sources relevant for education, employment and skills policy, outlining for each their practical applications, strengths and limitations, and the types of conclusions that can legitimately be drawn. This helps decision makers select appropriate evidence for different policy questions and avoid common misinterpretation.

The paper also examines what is at stake. Better data can enable sharper financial decision making – supporting cost-benefit analysis, revealing fiscal inefficiencies and improving the targeting of resources towards high-return programmes – while reinforcing the already well-established economic and social returns to skills investment. At the same time, governments are not the sole actors in skills development: employers, private providers and individuals invest significantly in training, and better public data should complement rather than substitute for these contributions.

To understand what prevents progress, the paper introduces a framework grouping the main obstacles into four categories: institutional, governance and financing barriers (such as fragmented mandates and unstable funding for data infrastructure); human-capital and analytical-capacity gaps (shortages of specialised data talent and limited data literacy among decision makers); legal and regulatory constraints (particularly around data protection and cross-agency sharing); and technical and interoperability challenges (legacy systems, inconsistent standards and cybersecurity risks). Examples from across the OECD illustrate how governments are addressing these barriers through integrated data systems, centralised analytical units and legal reforms that enable secure data linkage.

The paper concludes that improving data systems is not primarily about collecting more information. In many countries, substantial relevant data already exist but remain underutilised because they are fragmented, stored in incompatible systems or inaccessible to decision makers. The priority is to unlock and integrate existing data assets through better co-ordination, stronger analytical capacity and governance arrangements that encourage evidence use. While the costs of modernising data infrastructure are real, the potential gains – in more effective resource allocation, improved policy outcomes and stronger public trust – are considerable. Digital technologies, including artificial intelligence, can act as multipliers of these returns, but only when supported by strong data governance and high-quality underlying data.

1 Introduction

Public investment in skills – used here broadly to encompass the knowledge, competencies and qualifications acquired through education and training, from early childhood to adult learning – is a cornerstone of economic and social development. Higher levels of education and training are consistently associated with better employment outcomes, higher earnings, and stronger social cohesion, though returns vary across pathways, population groups and context. Governments thus see skills spending not merely as a recurrent cost but as an investment in future prosperity.

Approaches to financing skills development vary considerably across countries and stages of the life course. Each system combines a particular mix of financial instruments – ranging from supply-side funding of institutions to demand-side mechanisms such as subsidies or vouchers – shaped by national priorities and institutional contexts. The effectiveness of these arrangements depends on their alignment with policy goals and on the quality of the underlying evidence base. Efficient resource allocation has become increasingly important, not only in terms of how much is spent but in ensuring that funds are directed to interventions with proven impact.

At the same time, governments operate under growing fiscal constraints. Ageing populations, rising health and social care costs, climate adaptation, and new security pressures continue to exert pressure on public budgets. In this constrained environment, skills investment must compete with other critical priorities. Redirecting resources from inefficient or low-impact programmes proves politically challenging, as entrenched interests and institutional inertia resist change. However, tighter fiscal conditions make evidence-based decision making indispensable. It enables governments to justify, sustain and, where appropriate, expand investment in skills development by demonstrating clear value for money and measurable outcomes.

1.1. Why better data are essential

Across the OECD, the lack of timely, reliable and integrated data is a major barrier to strategic decision making in skills investment. Governments often lack comprehensive information on the effectiveness of education and training programmes, the needs of different population groups, and the long-term labour market outcomes of participants. While relevant data sources are often available – from administrative records and employer surveys to tax and employment registers – their coverage, quality and accessibility vary significantly across countries. In many cases, data remain fragmented, incomplete, or difficult to access and use for policy purposes.

OECD work on the data-driven public sector shows that these challenges are not unique to skills policy. Governments across policy domains face similar difficulties, including uneven data quality, limited internal data sharing, and barriers to interoperability that prevent data from being used as a strategic asset for policymaking and service improvement (OECD, 2019^[1]). However, these challenges are particularly acute in skills policy. Skills systems span multiple ministries, levels of government and stakeholders, from education and training providers to employers and social partners. This institutional complexity makes it more difficult to assemble the information needed to assess outcomes across the life course and to link spending with results. OECD analysis also highlights the breadth and diversity of stakeholder engagement

arrangements in skills systems, further underscoring the co-ordination challenges involved (Reznikova, Labanino and McKee Mathews, 2024^[2]). As a result, gaps in data integration and use have more direct consequences for the allocation and effectiveness of public investment in skills.

Without quality data, governments cannot determine whether skills investments are generating the intended outcomes or how to target resources for maximum effect. This leaves them vulnerable to inefficiencies and missed opportunities. Funds may flow to programmes with weak results – even where this reflects deliberate policy choices to support disadvantaged groups – while high-impact initiatives remain underfunded. Insufficient data have thus become a critical bottleneck, preventing the alignment of skills policies with economic and social needs.

It is important to recognise that governments are not the sole actors in developing and deploying skills. Employers, private providers and individuals invest significantly in training, often responding more rapidly to evolving labour market demands than public systems can. Better public data should complement – not substitute for – these private contributions.

Better data would transform policymaking. Consider the questions governments currently struggle to answer. Which education and training pathways deliver lasting employment gains? Which sectors and regions generate the highest returns from upskilling? How do different forms of financing affect participation and outcomes across the life course and across different population groups, particularly those at risk of exclusion? With reliable evidence, governments can make more strategic choices. They can invest where impact is greatest, adjust programmes to meet evolving labour market demands, and ensure resources are used efficiently, effectively and equitably – goals that are complementary rather than competing in the context of public skills financing. Advances in digital tools, including artificial intelligence, further expand these possibilities by enabling more granular analysis, improved forecasting of skills needs and more responsive allocation of funding. However, their effectiveness depends critically on the availability of high-quality, integrated data.

1.2. Purpose and scope

While closing data gaps has significant potential to strengthen policy decision making, doing so requires overcoming a set of structural, institutional and technical barriers. These challenges are diverse and operate at different levels of the system, shaping how data are collected, governed, shared and used in practice. To support a clearer and more action-oriented understanding, this paper proposes a framework that groups the barriers into four categories: institutional, governance and financing barriers; human-capital and analytical-capacity gaps; legal and regulatory constraints; and technical and interoperability challenges. Figure 1 illustrates these four categories, summarising what each looks like in practice and why it matters for decision making and the financing of skills systems. Each category involves distinct implications for policy design and requires different levers for reform. By structuring the analysis in this way, the section aims to provide a practical basis for identifying priorities and sequencing actions that enable progress toward more integrated and effective skills data systems. Throughout this paper, “skills” refers broadly to the knowledge, competencies and qualifications acquired through education and training, consistent with its use in international policy discourse.

This policy paper examines how stronger data on skills outcomes and investments can enable smarter public spending decisions, and what obstacles currently impede this goal. While many of the data issues discussed are relevant to skills policy more broadly, the paper’s focus is on data insofar as they inform financing and resource-allocation decisions: assessing programme effectiveness, targeting investment and linking spending to outcomes. Section 2 identifies current data gaps in education and skills policy and the consequences of those gaps for decision making. Section 3 provides an overview of the main data sources available for education, employment and skills policy, outlining what each source can and cannot support in terms of evidence and decision making. Section 4 discusses what is at stake – the potential

gains in efficiency and socio-economic outcomes if these gaps are addressed. Section 5 analyses the main barriers to progress, including institutional, governance and financing constraints, legal and regulatory challenges, capacity limitations and technical interoperability issues. Finally, Section 6 presents the conclusions.

Figure 1. A framework for understanding barriers to integrated skills data systems



Note: The four barrier categories are not mutually exclusive. Effective reform typically requires action across multiple dimensions simultaneously.

2 Current data gaps and their consequences

Recent decades have seen remarkable progress in data collection across OECD countries. Governments now routinely gather information on enrolment, graduation rates, and basic labour market outcomes. Administrative systems have become more sophisticated, and statistical agencies have expanded their capacity to monitor education and training. Yet despite these advances, significant gaps remain in the evidence available to guide skills policy.

Many countries still lack a unified and comprehensive view of their skills systems. Information on education, training, and labour market outcomes is often fragmented across multiple agencies and databases. Where data do exist, they are frequently incomplete, inconsistent, or incomparable across time and place.

The problem goes beyond the availability of basic statistics. One critical gap concerns financial information. While governments track overall education budgets, they often lack detailed data on how much is spent on specific programmes, and how spending is distributed across different population groups and stages of the life course. Understanding the flow of public resources is essential, but it represents only one piece of the puzzle. This challenge is closely linked to the absence or weakness of quality assurance mechanisms. Evidence from OECD work on adult education and training shows that increased spending, in the absence of robust quality assurance and outcome tracking, can lead to the expansion of low-quality provision and weak returns on investment (OECD, 2024^[3]).

Policymakers need a broader set of indicators to understand whether investments are achieving their goals. They need data on participation and completion rates, on how graduates perform in the labour market, on the quality and relevance of training, and on equity across different groups and regions. With these complementary indicators, policymakers can better see the full picture. They can track how skills are developed and used across the life course, or judge whether public resources are being deployed effectively.

This section identifies the main dimensions of these data gaps, highlights which policy areas are most affected, and explains how they hinder effective decision making.

2.1. Fragmentation of data systems

In many countries, skills-related data are dispersed across siloed systems managed by different ministries, agencies and levels of government (OECD, 2019^[4]). This fragmentation typically reflects the division of responsibilities across government portfolios and across levels of administration. In some systems, national governments set policy while subnational authorities manage delivery. In others, particularly federal systems, key functions are allocated across levels. For example, education is largely a state responsibility in the United States and a provincial responsibility in Canada.

Data systems are typically developed by different public bodies, each aligned with their specific mandate and regulatory requirements. As a result, information is collected and managed in separate systems reflecting sectoral responsibilities – for example, education authorities focus on enrolment and graduation,

labour ministries on employment and training programmes, and tax and social security institutions on earnings and employment histories. While these systems serve important administrative functions, they are rarely designed to operate together. Fragmentation is not only institutional or technical but also semantic: systems may be connected yet remain difficult to use jointly if they rely on different definitions, nomenclatures, metadata or exchange standards (see Section 2.4).

This limits their comparability and integration, making it difficult to link information across education, training and labour market outcomes. As a result, policymakers lack a consolidated view of the skills system and face constraints in assessing how individuals move through it and how public spending translates into results.

The consequences for policy are severe (see Section 3). Fragmentation limits the ability to trace how investments in education and training translate into labour market outcomes. Without linking school or training records to employment and earnings data, policymakers cannot answer basic questions. How do graduates fare in the job market? Which qualifications lead to sustainable employment? Which training programmes deliver the best returns? Similarly, budget data on skills programmes may not be connected to outcome indicators, making it difficult to assess value for money.

A further complication is timescale: tracing the link between education investments and sustainable employment outcomes can take years or even decades, by which point the economic and occupational landscape may have shifted substantially. The result is that governments often work with partial and disconnected information. This hampers their capacity to design coherent strategies or to allocate resources where they are most effective. Policy decisions are inevitably made with incomplete evidence, but the degree of incompleteness matters. Better data can narrow the margin of uncertainty, even if it cannot eliminate it. Without integration, however, opportunities to learn from past investments are lost.

2.2. Lack of longitudinal linkages

A second major gap concerns the limited availability of longitudinal data that track individuals over time. Understanding the impact of skills investments requires following people through their education, training, and careers. Yet in many countries, this proves difficult. Education, employment and earnings data are collected separately, often with no unique identifiers or data-sharing arrangements that allow them to be matched. Individuals can be observed at discrete moments – when they enrol in a course, when they graduate, when they register as employed – but the connections between these moments are often difficult to establish. Their subsequent trajectories disappear from view. These linkages matter not only between education and work, but also within education systems themselves: tracking trajectories such as dropout, grade repetition, transitions between levels and access to higher education is essential for understanding where and why skills development falters. Some countries, notably the Nordic nations, have made significant progress in this area by linking education, employment and earnings records through unique personal identifiers – demonstrating that integrated longitudinal data systems are achievable. Australia's Person-Level Integrated Data Asset (PLIDA) similarly enables longitudinal tracking of individuals before and after training, combining multiple administrative sources to assess outcomes such as post-completion earnings. France's InserJeunes platform links school, apprenticeship and employment records to monitor, within a single framework, both further study and the labour market insertion of young people leaving vocational programmes.

The consequences are profound. Without longitudinal linkages, policymakers cannot assess the long-term effects of different education and training pathways. Critical questions go unanswered. Do graduates of vocational programmes secure stable jobs? How do their earnings evolve over the course of their careers? Does mid-career training improve employment resilience during economic downturns? Evaluating the effectiveness of a training programme requires knowing not only who participated, but also how their employment outcomes changed compared to similar individuals who did not take part. In the absence of

linked data, such analysis is rarely possible. Notable exceptions exist, particularly among countries that have invested in integrated administrative data infrastructures.

The absence of longitudinal evidence also prevents governments from identifying which groups benefit most from training and which may need additional support. Without longitudinal data, such disparities remain hidden. Policies risk overlooking those who are less well served, and resources may flow to groups who would have succeeded regardless of intervention.

Ultimately, without these linkages, skills policies are designed on short-term snapshots rather than a fuller picture of outcomes. At the same time, even comprehensive life-course data must be interpreted with caution, as the labour market conditions under which outcomes were observed may differ from those facing current cohorts.

2.3. Coverage gaps across the skills system

Data collection is generally strongest for formal education systems. Primary, secondary, and tertiary education are well documented in most OECD countries, with regular reporting on enrolment, completion, and basic outcomes. Yet coverage weakens at both ends of the learning continuum. Early childhood education and adult training remain poorly monitored, despite their growing importance for skills development across the life course.

In early childhood, data collection is particularly fragmented and patchy. Early childhood services often operate under multiple regulatory frameworks (Leseman and Slot, 2025^[5]; OECD, 2019^[6]). Some providers fall under the purview of education ministries, while others are overseen by social services, family affairs, or health ministries. Private providers, community organisations, and informal care arrangements further complicate the landscape. This institutional fragmentation makes comprehensive data collection challenging. Many countries lack consistent information on participation rates, quality of provision, staff qualifications, and developmental outcomes across these diverse settings. Data collection often begins only when children enter primary school, leaving the crucial first years largely undocumented. This creates difficulties in evaluating the impact of early investments on later educational performance and labour market success. The gap is particularly problematic given the well-documented importance of strong foundations in the early years and the long-term returns that early interventions can generate.

Adult learning represents an even larger blind spot in skills data. Many adults acquire new competences through non-formal courses, employer-provided training, apprenticeships, or online platforms. Yet these activities are rarely captured in official statistics. Because much of this learning occurs outside formal institutions and across diverse settings, systematic monitoring of participation, investment, and outcomes is limited (OECD, 2025^[7]). Employer-sponsored training, which represents a substantial share of adult learning activity, is particularly difficult to track. One practical reason is that firms rarely structure their accounts in ways that allow training-related costs to be readily identified, making accurate reporting burdensome and potentially unreliable. As a result, governments have little visibility on how much employers invest in training, which workers participate, what skills are being developed, and what benefits these efforts generate over time. In many cases, employers address evolving skills needs directly through in-house training, often without reference to government-collected data or public policy frameworks. This underscores the importance of understanding, rather than supplanting, private training activity.

The consequences are significant. Policies to expand early childhood education or lifelong learning are often introduced with incomplete evidence on who participates, the quality of provision, and the outcomes achieved. Without such data, it is difficult to know whether disadvantaged families have equitable access to high-quality childcare, or whether mid-career reskilling programmes genuinely help workers transition into stable employment. This lack of visibility means that vulnerable groups may be underserved without policymakers realising it. Children from low-income households may miss out on early interventions that

could transform their trajectories. Adults in low-skilled jobs, older workers, and those in rural or remote areas may have limited access to training opportunities. As a result, investments risk being directed towards those already better served, reinforcing rather than reducing inequalities.

2.4. Inconsistent definitions and quality issues

Even where data exist, differences in definitions, methodologies, and quality standards often limit their usefulness. Education and training data are not always collected using consistent classifications, making it difficult to compare across programmes, institutions, or regions. What one country records as vocational training may differ substantially from another, reflecting variations in programme duration, entry requirements, certification standards, and the balance between classroom instruction and workplace learning. Yet these differences may not be clearly reflected in the data, creating apparent comparability where little actually exists. These challenges are further magnified in international comparisons, where differences in definitions and data collection practices can lead to misleading conclusions if not carefully accounted for. The problem extends within countries as well. Regional authorities may apply different criteria for categorising participants or measuring completion, while institutions may report outcomes at different intervals or use different definitions of employment success.

These definitional inconsistencies are compounded by weak data quality, which further undermines usability. Errors in administrative records, outdated information, and missing variables all reduce the reliability of analyses. Consider earnings data that lack details on occupation, industry, or hours worked. Without this context, it becomes impossible to distinguish between full-time stable employment and precarious part-time work. Similarly, training records may note participation but not completion, or record completion without capturing the credential obtained. Demographic information is often incomplete, preventing analysis of equity across different groups. When datasets from different sources cannot be meaningfully combined due to incompatible standards and quality controls, the result is a fragmented system that resists integration despite the abundance of available information. Improving data quality requires practical instruments rather than general principles alone: data dictionaries, common schemas, harmonised reference values, clear data-entry rules and quality indicators can make data quality visible, measurable and improvable over time.

The consequence is a troubling paradox. Governments may find themselves unable to answer basic questions despite sitting on vast administrative databases, simply because the information they hold is too fragmented, inconsistent, or unreliable to yield meaningful insights.

2.5. Timeliness and accessibility

Another common limitation is that data are not available when they are most needed, creating a temporal mismatch between information collection and policy decisions. In many countries, there is a substantial lag between the collection of information and its release, sometimes stretching across multiple years. Administrative data on training programme participation, completion and employment outcomes, for instance, may take one or two years to be processed, validated, and published, even though the raw information exists in government systems much earlier. By the time these data become available, budget allocations have been set and programme adjustments have been made based on older evidence or anecdotal information.

Accessibility poses an additional challenge. Even when relevant data are available, legal restrictions, confidentiality rules, or technical limitations often prevent policymakers and analysts from accessing them. While these safeguards serve essential purposes in protecting individual privacy (see Section 5.3), their cumulative effect can limit the analytical use of data for policy purposes. In many cases, policymakers must

rely on outdated or less relevant indicators simply because more timely and detailed data are not accessible.

The combined effect of these delays and access restrictions is profound. Decisions are frequently made on the basis of incomplete or outdated evidence, undermining governments' ability to respond swiftly to economic shocks or adapt training systems to evolving skills demands. Emerging inequities may go undetected until they have already widened significantly.

2.6. Granularity: Equity and regional blind spots

Insufficient granularity in data can obscure important disparities across population groups and regions. In many countries, data are collected at a highly aggregated level, with limited disaggregation by gender, age, region or rurality, socio-economic background, migration status, disability status, or ethnic, cultural or linguistic background. Without this level of detail, inequalities in access to and outcomes from training may go undetected. National averages may show strong overall employment outcomes after vocational training, while concealing that women or disadvantaged minorities benefit far less.

Regional differences often remain similarly hidden. Subnational data on participation, investment, and outcomes are frequently incomplete or inconsistent, making it difficult to assess how national policies perform in different local contexts. A region with low training participation and high unemployment may be masked by national figures, and varying regional data standards can further hinder comparison across jurisdictions. More granular data would also support territorial planning processes, helping to identify differentiated skills needs across regions, productive sectors and local demographic dynamics.

The lack of granular information has direct implications for policy. Governments may inadvertently allocate resources to groups or areas that are already well served, while those most in need remain overlooked. Workers in remote areas, for example, may have limited access to upskilling programmes, and individuals from low-income households may face barriers to participation that go unaddressed. Limited granularity therefore undermines both the efficiency and the equity of skills policies.

3

Types of data sources for decision making, and what they can and cannot tell policymakers

Effective policy design, implementation and evaluation require a clear understanding of the different types of data that inform decisions across education, employment and skills systems. Multiple sources of data exist, each offering distinct strengths and responding to different analytical needs. Table 1 summarises these sources, their main uses, and their key strengths and limitations. Figure 2 complements this overview by presenting a heat map that visually compares the relative strengths of each data source across key analytical dimensions, including coverage, timeliness, granularity and capacity for causal inference.

A challenge that can arise in policymaking is the expectation that available datasets can answer a wider range of policy questions than their design allows. In practice, the ability to draw valid conclusions depends on both the analytical capacity of decision makers and the structure and quality of the underlying data. Some sources are appropriate for describing trends, diagnosing disparities and informing strategic planning, while others are required for assessing long-term trajectories or determining whether a policy or programme caused an observed change. Misinterpretation of data – such as drawing causal conclusions from descriptive indicators or extrapolating results beyond the population studied – can lead to weak or misleading policy choices.

For this reason, policymakers need a clear understanding of the kinds of questions that different data sources can answer reliably, and those for which they are not suitable. This includes knowing which can support system-level monitoring and allocation decisions, which can credibly inform programme evaluation, and which require complementary evidence to avoid incorrect inference. Recognising the analytical boundaries of each dataset is as important as understanding its potential.

Ten data sources are relevant for education and skills policy, organised into five broad categories that reflect distinct analytical logics and policy functions: population and household surveys, administrative and assessment data, programme-focussed data, evaluative and qualitative methods, and emerging data sources. While these categories are not mutually exclusive, recognising the distinct logic of each helps policymakers identify which sources are fit for purpose for a given analytical need, and which combinations of evidence are required to support robust decisions. Table 1 summarises these ten sources within their respective categories, outlining their main uses for policymaking and their key strengths and limitations. Figure 2 complements this overview with a heat map comparing the relative strengths of each source across key analytical dimensions, including coverage, timeliness, granularity and capacity for causal inference. Annex A provides a detailed profile of each data source, including illustrative examples, and the types of conclusions that can legitimately be drawn from each.

Table 1. Summary table: Data sources for decision making

Data source	Main uses for policymaking	Strengths	Limitations
Cross-sectional surveys and Labour Force Surveys			
1. Cross-sectional household and individual surveys	Mapping population characteristics and inequalities; monitoring participation and attainment	Representative; comparable over time; broad population coverage	Cannot track trajectories; limited causal inference; self-report bias
2. Labour Force Surveys (LFS)	Diagnosing labour market conditions; workforce planning; identifying vulnerable groups and sector trends	High frequency; strong trend analysis; internationally comparable	Limited programme-level insight; sensitive to economic cycles; weak explanatory power
Administrative records and standardised assessments			
3. Administrative data	Real-time monitoring; resource allocation; identifying service gaps; linking records over time	Large-scale or full population coverage; granular detail; high frequency	Collected for operations rather than analysis; inconsistent definitions across systems; limited causal inference on its own
4. Standardised assessments and skills surveys	Benchmarking performance; diagnosing system quality; monitoring equity and learning outcomes	Strong comparability; robust outcome measurement; identifies disparities clearly	Covers a limited set of skills; periodic rather than continuous; cannot explain underlying mechanisms on its own
Programme-focussed data			
5. Longitudinal and panel data	Tracking pathways and transitions; understanding long-term outcomes; supporting programme evaluation	Captures trajectories over time; supports stronger inference than cross-sectional data; medium- to long-term perspective	Costly to maintain; attrition risk; narrow scope; not inherently causal without appropriate evaluation design; raises privacy concerns when linking records across systems
6. Tracer studies	Assessing programme-specific outcomes; understanding school-to-work and training-to-work transitions; generating early implementation feedback	Focused on specific cohorts or programmes; practical outcome insights; useful for transition analysis	Often limited in representativeness; weak comparability across studies; counterfactual usually absent
7. Employer and enterprise surveys	Identifying skill shortages and recruitment challenges; informing training supply; supporting sector planning	Direct employer perspective; detects emerging demand; relevant for sectoral analysis	Perception-based responses; firm-size and sector bias; weak cross-country and cross-survey comparability
Evaluative and qualitative methods			
8. Experimental and pilot programme data	Testing intervention effectiveness; informing decisions on scale-up, redesign or discontinuation	Strong internal validity when well designed; strongest basis for causal inference; supports cost-effectiveness analysis	Often small-scale; limited external validity; costly and time-intensive
9. Qualitative data and case studies	Understanding mechanisms, behaviours and implementation barriers; refining policy and programme design	Rich contextual insight; captures lived experience; helps explain why policies succeed or fail	Not designed for population-level inference; limited comparability; findings depend on case selection and interpretation
Emerging data sources			
10. Big data, learning analytics and online job postings	Detecting emerging skill needs; real-time monitoring of demand signals; supporting rapid adjustment	Timely; granular; useful for horizon scanning and near real-time intelligence	Platform and selection bias; variable data quality; incomplete coverage of informal sectors and occupations; no causal inference

Figure 2. Analytical strengths of data sources for education and skills policymaking

	POPULATION COVERAGE	TIMELINESS & FREQUENCY	GRANULARITY & DETAIL	CROSS- NATIONAL COMPARABILITY	CAUSAL INFERENCE	TRAJECTORY ANALYSIS	PROGRAMME EVALUATION	LABOUR MARKET INSIGHT
POPULATION & HOUSEHOLD SURVEYS								
1 Cross-sectional surveys	STRONG	MODERATE	MODERATE	STRONG	LIMITED	NONE	LIMITED	MODERATE
2 Labour Force Surveys (LFS)	STRONG	STRONG	MODERATE	STRONG	LIMITED	LIMITED	LIMITED	STRONG
ADMINISTRATIVE & ASSESSMENT DATA								
3 Administrative data	STRONG	STRONG	STRONG	LIMITED	LIMITED	MODERATE	LIMITED	LIMITED
4 Standardised assessments	MODERATE	LIMITED	MODERATE	STRONG	LIMITED	LIMITED	LIMITED	MODERATE
PROGRAMME-FOCUSED DATA								
5 Longitudinal / panel data	MODERATE	LIMITED	STRONG	MODERATE	MODERATE	STRONG	MODERATE	MODERATE
6 Tracer studies	LIMITED	MODERATE	MODERATE	LIMITED	LIMITED	LIMITED	MODERATE	MODERATE
7 Employer surveys	MODERATE	MODERATE	MODERATE	LIMITED	NONE	LIMITED	LIMITED	STRONG
EVALUATIVE & QUALITATIVE METHODS								
8 Experimental & pilots	LIMITED	LIMITED	STRONG	LIMITED	STRONG	MODERATE	STRONG	MODERATE
9 Qualitative research	LIMITED	MODERATE	STRONG	LIMITED	LIMITED	LIMITED	MODERATE	LIMITED
EMERGING DATA SOURCES								
10 Big data & job postings	MODERATE	STRONG	STRONG	MODERATE	NONE	LIMITED	NONE	STRONG

STRENGTH: None / Very weak Limited Moderate Strong

Note: Each cell indicates how well a data source serves a given analytical purpose.

4 What is at stake: Potential gains from closing the gaps

Strengthening the availability, quality and use of data in skills policy is not an abstract administrative exercise. The potential gains are concrete and considerable. Closing current data gaps and improving the effective use of diverse evidence sources could transform the way governments allocate resources, design interventions and evaluate results. Better data do more than refine analysis: they enable more rigorous financial decisions, improve system efficiency, enhance equity and support more responsive and accountable policymaking. This section outlines what is at stake – both the immediate benefits and the long-term gains that can be realised when decisions are grounded in robust evidence.

4.1. Sharper financial decision making: Reallocating to high-return programmes, avoiding low-yield spend, managing fiscal risk

Closing the data gaps outlined in Section 2 would enable more strategic and evidence-based financial decisions in skills policy. Improved data would allow governments to conduct systematic cost-benefit analyses and identify which programmes deliver the highest returns on investment. With such evidence, funding can be redirected toward more effective initiatives and away from those with limited impact. This helps overcome the inertia that often keeps budgets locked in historical patterns rather than aligned with demonstrated results. Reliable data provide the analytical basis and accountability needed to implement such reallocations transparently.

Better data would also strengthen governments' capacity to identify and address fiscal inefficiencies. In tight budget environments, investments that fail to deliver represent not only wasted resources but also missed opportunities. Timely monitoring data can act as an early warning mechanism, flagging underperforming programmes that require redesign or reallocation. For example, persistently low uptake of subsidised training vouchers in certain regions may signal flaws in programme design or communication. Moreover, robust longitudinal data allow for more accurate forecasting of future costs and benefits, reducing the risk of over-committing funds to initiatives that do not yield sustainable outcomes.

Enhanced data granularity further improves targeting and prioritisation. By identifying where marginal returns are highest, policymakers can allocate resources to groups, sectors, or regions where investment is likely to have the greatest impact. For instance, evidence may reveal that mid-career digital skills training in lagging regions produces stronger employability gains than marginal increases in funding for programmes that already serve well-connected populations – differences that are not always apparent without granular data. Data-driven targeting thus contributes to a more efficient and equitable allocation of public funds.

Finally, real-time and outcome-based data systems can enhance both agility and accountability in public spending. Governments can move toward funding approaches that are more responsive to evolving labour market signals – for example, adjusting training budgets as demand shifts across sectors – while recognising that labour markets are inherently complex and cannot be fully steered through data alone.

Performance monitoring frameworks linked to measurable outcomes can also strengthen accountability, ensuring that programme managers are incentivised to use public funds effectively.

In sum, closing data gaps has the potential to contribute to a virtuous cycle in which more evidence-based investments lead to stronger outcomes, though the empirical evidence linking data investments directly to aggregate employment, productivity and growth outcomes remains limited and context dependent.

4.2. Economic and social returns over the short and long term

Comprehensive and reliable data can strengthen the already well-established economic and social returns to investment in skills. Across OECD countries, tertiary graduates contribute more in taxes, depend less on social transfers, and report better health and well-being than those with lower attainment (OECD, 2025^[8]; OECD, 2021^[9]). These associations, however, partly reflect selection effects: individuals who attain higher education often benefit from pre-existing advantages that independently contribute to better outcomes. The returns to expanding tertiary access are therefore context-dependent. Even modest improvements in adult basic skills can be linked to higher employability, earnings, and productivity, while broader gains in literacy and digital proficiency support long-term growth (OECD, 2025^[10]). Returns also extend beyond employment and earnings: education and training generate benefits in terms of learning outcomes, civic participation, socio-emotional development and well-being, which data systems should aim to capture alongside labour market indicators.

Closing data gaps enables governments to allocate resources more effectively, directing investment toward programmes and populations that deliver the strongest outcomes. Integrated data systems allow for policies that build a more adaptable workforce, align training provision with labour-market demand, and expand opportunities for under-represented groups. Evidence-based allocation thus improves both the efficiency and the equity of skills spending.

It is useful to distinguish between the immediate and longer-term benefits of improving data use. In the short term, closing selected data gaps can deliver tangible efficiency gains by revealing obvious overlaps or inefficiencies. For example, an integrated data dashboard may quickly uncover duplicate training programmes funded by different agencies, allowing consolidation and immediate cost savings. Such early actions are typically low-cost, achievable within a single budget cycle, and demonstrate the practical value of better data, helping to build institutional momentum for broader reform.

The larger benefits, however, emerge over the longer term. Developing comprehensive data infrastructure and analytical capacity takes time – new graduate tracking systems, longitudinal studies, or inter-agency data links may require several years to become fully operational – but once established, they enable continuous optimisation of spending and policy design. While rigorous evidence directly linking investments in data infrastructure to improved policy outcomes remains scarce, the experience of countries with more integrated systems – such as the Nordic nations and Estonia, where linked education, tax and employment records and unified digital identities enable routine tracking of outcomes across policy domains – points to meaningful gains in policy responsiveness and analytical capacity. Over time, this fosters a culture of evidence-based improvement, in which the skills system evolves and adapts in response to changing needs. Each cohort benefiting from more effective investment contributes to sustained economic and social returns.

5

Barriers to building integrated skills data systems

Improving the quality, accessibility and use of data in education and skills policy is not only a technical task but an institutional and systemic challenge. Countries often face multiple, interconnected obstacles across institutions, legal frameworks, capacities and technical systems that affect how data are collected, shared and used for policymaking. To support a clearer and more operational understanding of these challenges, this paper introduces a simple framework that groups the key barriers into four categories: institutional, governance and financing barriers; human-capital and analytical-capacity gaps; legal and regulatory constraints; and technical and interoperability challenges. This framework is designed to help policymakers identify where the main bottlenecks lie, understand their implications for data use, and prioritise the levers required to strengthen integrated skills data systems. The following sections discuss each of these four barrier types in detail.

5.1. Institutional and governance barriers

5.1.1. Fragmentation and lack of co-ordination

Despite the potential of data to strengthen decision making, institutional and governance barriers often limit its effective use. A recurrent challenge is the fragmentation of mandates and responsibilities across the skills system. Education and training involve multiple ministries (education, labour, finance), several levels of government (national, regional, local) and a wide range of stakeholders, including public institutions, private providers and employers. In many countries, no single authority co-ordinates a unified data strategy, and responsibilities for collecting and managing information are dispersed across agencies. Labour ministries may manage unemployment and training data, education ministries oversee school and higher education records, and regional authorities collect information on vocational programmes, each operating in separate systems. This fragmentation creates duplication in some areas, gaps in others and, critically, prevents the development of a coherent picture of how skills are developed, used and updated across the life course. OECD analyses consistently point to weak co-ordination across ministries, levels of government and stakeholders as a major barrier to effective skills policy implementation (OECD, 2019^[4]).

Emerging tools, including artificial intelligence, can help map existing data assets, harmonise definitions and support shared analytical platforms across institutions. However, their effectiveness depends on clear governance arrangements and agreed standards; without these, technology risks reinforcing rather than reducing fragmentation.

5.1.2. Incentives, culture and financing constraints

Weak co-ordination is often reinforced by institutional cultures and incentive structures. Many administrations treat their datasets as proprietary resources, with limited trust in how other bodies may use the information. This reduces willingness to share data and limits the role of evidence in decision making. When data-driven practices are not embedded, leaders may rely on precedent or political judgement, and

analytical work receives limited attention. Developing a culture that values evidence requires clear leadership, predictable rules for data use and incentives that encourage collaboration across institutions.

Financing and sustainability issues further constrain progress. Many improvements to data systems begin as short-term projects funded through pilots or external grants. When these end, systems often lack stable resources for maintenance, upgrades or analytical capacity. Budgeting rules may prioritise visible new programmes over long-term data infrastructure, and efficiency gains identified through better data use may be clawed back, reducing incentives to improve. Without predictable funding and aligned incentives, the benefits of data investments are difficult to sustain.

5.1.3. Capacity, accountability and policy leadership

Institutional arrangements also influence how effectively data can support policy. Responsibilities for outcomes are often distributed across multiple ministries and levels of government, yet accountability frameworks do not reflect these interdependencies. When no institution is clearly responsible for results that depend on shared data, gaps emerge between information and action. For example, education authorities may not engage with employment data if labour market outcomes fall under another ministry, even though such information is essential for programme planning.

Addressing these challenges requires both structural and leadership-driven reforms. OECD analysis shows that strong governance arrangements and explicit co-ordination mechanisms are often decisive in shifting systems away from siloed practices and toward more integrated, outcome-oriented approaches (OECD, 2024^[11]). Effective governance also requires transparency about how data are produced – including collection methods, definitions, response rates and deviations from international standards – and clearly assigned responsibilities across the data lifecycle, from defining legitimate purposes and validating access to quality assurance, metadata maintenance and accountability for data reuse. Several OECD countries have taken steps to address these issues by creating central units or observatories with a mandate to integrate data across agencies and report to high-level bodies (see Box 1). Estonia's Education Information System (EHIS) links education, employment and population data under the oversight of the Ministry of Education and Research. Ireland's Skills and Labour Market Research Unit (SLMRU), hosted by SOLAS, provides integrated analysis to the National Skills Council. Norway's Statistics Norway uses linked administrative datasets to support cross-ministerial decision making and reports jointly to the Ministry of Education and the Ministry of Labour. Chile's Integrated System of Higher Education Information (SIES) consolidates education and labour data to inform national skills and financing policies. Australia's National Centre for Vocational Education Research (NCVER) acts as the national custodian of VET data, while Jobs and Skills Australia provides independent analysis of current and future skills and workforce needs. The effectiveness of such arrangements depends on their integration with line ministries and on clear mechanisms to ensure that analytical findings translate into operational decisions, rather than remaining disconnected from policy processes.

Box 1. Country example: Finland

Finland's Service Centre for Continuous Learning and Employment (SECLE)

Finland's Service Centre for Continuous Learning and Employment (SECLE) provides an example of how institutional arrangements can help overcome fragmentation in skills systems. The Centre operates under the joint oversight of the Ministry of Education and Culture and the Ministry of Economic Affairs and Employment, two key actors in skills policy. This dual governance structure reflects the cross-cutting nature of skills development and helps align policy objectives across education and labour market domains.

SECLE's mandate focusses on strengthening the skills of the working-age population and improving the match between labour market needs and workforce competencies. A central element of its approach is the development of integrated data capabilities. The Centre is investing in a comprehensive data infrastructure that links information on education, training and employment outcomes, combining existing administrative datasets with new digital tools and cloud-based solutions.

By creating a shared evidence base across institutions, SECLE supports more co-ordinated decision making and helps bridge gaps between policy design and labour market outcomes. As a relatively recent initiative, systematic evidence on SECLE's measurable impact on labour market outcomes is still emerging. This approach illustrates how governance reforms – through joint mandates, shared accountability and integrated data systems – can reduce fragmentation and improve the strategic use of data in skills policy.

Source: The Service Centre for Continuous Learning and Employment (2026^[12]), *About Us*, www.jotpa.fi/en/about-us; Solita (2026^[13]), *SECLE supports the employment market with data*, www.solita.fi/work/secle-supports-the-employment-market-with-data/.

5.2. Human-capital and analytical-capacity gaps

5.2.1. Shortage of specialised data talent

Even the most advanced data systems have limited value without the human capacity to interpret and act on the information they generate. Many OECD countries face a shortage of data engineers, analysts and scientists in the education and skills domain (OECD, 2023^[14]) (OECD, 2024^[15]). These specialists are in high demand across all sectors, and public administrations often struggle to attract and retain them due to less competitive salaries, limited career progression, and more rigid human resource frameworks (OECD Observatory of Public Sector Innovation, n.d.^[16]).

This talent gap is particularly evident in advanced analytics. In many ministries of education or labour departments, only a small number of statisticians or data officers are responsible for managing large, complex datasets. Limited in-house capacity restricts the ability to conduct deeper analysis, increasing reliance on external consultants. While outsourcing can fill immediate needs, it is costly and risks fragmenting knowledge and reducing continuity within administrations. Emerging tools, including artificial intelligence, can help alleviate some of these constraints by automating routine data processing, supporting data linkage and enabling non-specialists to generate basic analysis. However, they do not substitute for analytical expertise and may increase the need for skills related to interpreting results, assessing model limitations and ensuring responsible use.

Talent retention is an additional challenge. The cost of competitive remuneration for such profiles is substantial and must be factored into the overall investment case for data system development. Even when

governments succeed in recruiting skilled data analysts, retaining them can be difficult if career prospects and remuneration are more competitive in the private sector. OECD analysis has underlined that the scarcity of digital talent, particularly in areas such as data science, remains a significant barrier to digital transformation and that public sector organisations need to better integrate these profiles and methods into their daily operations (OECD Observatory of Public Sector Innovation, n.d.^[16]).

Finally, continuous training for existing staff is another area of concern. Many officials responsible for managing education and labour market data have backgrounds in traditional statistics or IT, but may not be familiar with more recent methods of data analytics or visualisation. Without systematic professional development, staff skills risk becoming outdated. Targeted training in the use of new data tools, or in the interpretation of advanced analyses such as value-added models of school performance, could substantially increase effectiveness. However, in many administrations, training opportunities and budgets remain limited, constraining the capacity to modernise data practices. Box 2 provides examples of how countries are developing structured approaches to strengthen digital and data skills across the public sector. While these initiatives represent promising approaches, systematic evidence on their cost-effectiveness and measurable impact on public sector performance is still limited.

Box 2. Country examples: Korea, Lithuania and Canada

Korea: National Human Resources Development Institute (NHI)

The Korean NHI, established in 2016 under the Ministry of Personnel Management, provides training and learning opportunities for civil servants across government functions. Its programmes combine e-learning through the Talent Development Platform with regular curriculum courses.

A distinctive feature of the digital platform is its ability to track learning history, behaviour, and job profiles, allowing for personalised training recommendations based on individual skill levels.

The training primarily focusses on ICT fundamentals, artificial intelligence, and big data technologies, with an emphasis on their applications in government.

Source: Ministry of Personnel Management (2023^[17]), *Talent Development Platform*, www.learning.go.kr/intro/board/viewPlatformIntroduce.do; Burtscher, M., S. Piano and B. Welby (2024^[18]), "Developing skills for digital government: A review of good practices across OECD governments", <https://doi.org/10.1787/f4dab2e9-en>.

Canada: Canada School of Public Service (CSPS) Digital Academy

The CSPS was founded in 2018 to help civil servants build the skills needed for Canada's digital transformation. It provides a structured framework for developing key competencies and offers training through live digital courses, self-paced modules, and mandatory programmes. These are complemented by informal opportunities such as learning events and online resources, making training accessible to over 350 000 federal civil servants.

The Digital Academy covers ten areas: digital government, product management, AI, digital leadership, agile, data, emerging technologies, service design, cybersecurity, and cloud. In data-related topics, courses include data visualisation, navigating large databases, and using data exploration tools.

Housed within the government's School of Public Service, the Academy has a clear institutional mandate to drive skills development across the public sector.

Source: Government of Canada (2023^[19]), *CSPS Digital Academy*, <https://doi.org/10.1787/f4dab2e9-en>; Burtscher, M., S. Piano and B. Welby (2024^[18]), "Developing skills for digital government: A review of good practices across OECD governments", <https://doi.org/10.1787/f4dab2e9-en>.

Lithuania: GovTech Lab

Launched in 2019 as part of Innovation Agency Lithuania, GovTech Lab fosters public sector innovation by equipping organisations with tools, methods, and resources to solve challenges using technology. Its main activities include:

- **GovTech Challenges:** Co-ordinating structured co-creation programmes between the public sector and startups to develop pilot solutions.
- **Innovation Skills:** Offering workshops and study visits to enhance public sector innovation
- **Ecosystem and Community Building:** Running international networks and programmes

Source: GovTech Lab (2024^[20]) *What is GovTech Labs*, <https://govtechlab.lt/about/>; OECD (2024^[21]), *Enabling Digital Innovation in Government: The OECD GovTech Policy Framework*, <https://doi.org/10.1787/a51eb9b2-en>.

5.2.2. Data literacy and organisational barriers

Beyond technical specialists, a broader gap in data literacy persists among leadership and staff in many education and labour administrations. Senior decision makers – such as ministers, directors, or school leaders – often have limited training in interpreting statistical evidence. Where top officials are not comfortable with data, they may be less inclined to request rigorous analysis or to rely on it in decision making, even when robust information is available. This can reinforce reliance on intuition or political considerations. Building a culture of data-informed leadership is therefore essential. OECD research on digital government highlights that strengthening the skill mix in public administration is a priority – encompassing both digital competences and what might be termed “data skills for leadership” – the ability of senior decision makers to interpret evidence, ask the right questions of data, recognise its limitations, distinguish correlation from causal evidence, and interpret AI-generated outputs with appropriate caution (van Ooijen, Ubaldi and Welby, 2019^[22]).

Furthermore, organisational arrangements in some education systems do not always enable data analysts to contribute effectively. Analysts may be concentrated in small, under-resourced units, distant from policymakers and with limited influence in decision making processes. This can result in the under-utilisation of existing talent. More integrated approaches – such as embedding analysts within policy teams or creating multidisciplinary project groups – can strengthen the relevance of analysis while also improving job satisfaction and retention. This approach is not incompatible with the centralised units or observatories discussed earlier: centralised bodies can provide shared infrastructure and standards, while embedded analysts ensure that findings are applied in day-to-day policy work. Analytical capacity also depends on actors beyond government: an active research community provides an independent source of analysis of public data assets. Where this community is small or poorly connected to available data, valuable datasets risk remaining underutilised.

5.3. Legal and regulatory constraints

5.3.1. Data protection, consent and intellectual property

Legal and regulatory frameworks can shape, and at times constrain, the sharing and use of data for skills policy. Data protection and privacy regulations – such as those introduced under the General Data Protection Regulation (GDPR) – play a critical role in safeguarding personal information, including student records, training participation data and employment histories. However, these frameworks can also make it challenging to link datasets across systems or to use data for evaluation without explicit consent. For example, an education ministry seeking to follow graduates into the labour market by linking education

records with tax or social security data may face legal barriers, requiring individual consent or specific exemptions. In many countries, legislation does not explicitly provide for such data matching for policy purposes, leaving potentially valuable datasets disconnected. International evidence further highlights gaps in this area: a recent report found that only 16% of countries explicitly guarantee data privacy in education law, suggesting that in many jurisdictions data protection is either weak – undermining trust – or governed by general laws not tailored to education, creating uncertainty and caution among data holders (UNESCO, 2023^[23]).

Consent requirements can also present challenges. When explicit individual consent is required each time data is repurposed for analysis, projects can become impractical. Some countries have sought to address this by enabling the use of anonymised or pseudonymised data for research under strict safeguards, though not all legal regimes allow this without complex approval processes. Intellectual property and licensing issues may add further complications (OECD, 2019^[1]). For instance, standardised test data or curriculum materials may be owned by private entities, restricting their sharing or publication. Similarly, private learning platforms may decline to share data with governments, citing intellectual property rights or competitive sensitivities.

5.3.2. Cross-agency and cross-border data sharing

Legal barriers can also arise in cross-agency data sharing. In many administrations, laws or regulations restrict the use of data to the sector in which it was collected – for example, education data may only be used for educational purposes, and employment data only for employment services. Even where such restrictions are not explicit, legal ambiguity can create a “chilling effect”, with public officials reluctant to share data for fear of breaching rules. Without clear legal provisions, for example, a labour ministry may be advised not to share training outcomes with the education ministry. Some countries have sought to overcome these obstacles by enacting specific data-sharing legislation or inter-ministerial agreements that authorise integration for defined purposes, such as monitoring skill outcomes. Where such frameworks are absent, uncertainty and administrative procedures can delay or prevent progress in linking data.

Restrictions on transferring data across jurisdictions present an additional challenge, particularly for international comparisons or collaborative initiatives. Policymakers and researchers may wish to combine data from multiple OECD countries to benchmark performance or identify good practices. However, data localisation requirements and differing privacy regimes often complicate such efforts. Under the GDPR, for example, personal data generally cannot be transferred outside the EU to jurisdictions without equivalent safeguards, which limits sharing with partners in other regions unless the data is anonymised. Even within federations or regional groupings, variations in legislation can impede the flow of skills data.

Public sector data is also subject to freedom of information requirements and confidentiality provisions, which can at times be in tension. Open data initiatives encourage governments to make datasets accessible, while individual-level records on learners and trainees are confidential and require protection. Striking the right balance can be difficult, and in cases of uncertainty, the default response is often to restrict sharing.

5.3.3. Balancing privacy, utility and reform needs

Legal and regulatory frameworks serve a dual purpose: they are essential to protecting citizens’ rights and data security – which is fundamental for public trust – but if outdated or overly restrictive, they can also hinder the effective use of data for policy purposes. The challenge lies in enabling data sharing that is as secure as practicable and lawful, recognising that no system is entirely immune to breaches. Potential approaches include strengthening anonymisation and pseudonymisation techniques, creating legal “safe harbours” for the use of data in statistical and policy evaluation, and updating regulations to reflect the benefits of data integration. OECD work has emphasised the importance of adapting national statistical

laws to permit the secure linkage of education, employment and social data for approved research that serves the public interest, under strict privacy safeguards (OECD, 2019^[24]). Legal clarity is critical: when agencies understand precisely what is permitted, they can act with confidence rather than defaulting to inaction. Striking the right balance between privacy and utility – ensuring that data are not misused, while avoiding unnecessary barriers to its productive use – often requires legal reform or new regulations. Progress in this area can be slow, which is why legal frameworks continue to represent a major barrier category for effective data use. New Zealand’s Integrated Data Infrastructure (IDI) provides an example of this approach (see Box 3), enabling secure linkage of anonymised data across sectors to support policy analysis while protecting individual privacy.

Box 3. Country example: New Zealand

New Zealand: Integrated Data Infrastructure (IDI)

The Integrated Data Infrastructure (IDI) is a large, anonymised longitudinal microdata resource that links information from multiple sources, including government agencies, official New Zealand surveys, and non-government organisations (NGOs), among others.

By excluding personal identifiers such as names, dates of birth, and identification numbers, the IDI enables cross-sector research while safeguarding individual privacy. This makes it a powerful tool for evidence-based policy development.

The IDI encompasses eight main categories of data: Health; Education and Training; Benefits and Social Services; Justice; People and Communities; Population; Income and Work; and Housing.

Source: Stats NZ Tatauranga Aotearoa (2022^[25]), *Integrated Data Infrastructure*, www.stats.govt.nz/integrated-data/integrated-data-infrastructure/.

5.4. Technical and interoperability challenges

5.4.1. Legacy systems

Technical barriers remain a major constraint on the effective use of data in education and skills policy. A key challenge lies in the persistence of legacy IT systems. Many public-sector databases were developed decades ago on mainframe technologies designed for stability but not for interoperability. These systems are costly to maintain, rigid, and difficult to connect with modern applications. By contrast, newer platforms – often based on SQL databases or cloud architectures – allow modular design, easier updates and real-time data sharing.

This mismatch makes interoperability highly complex. For example, linking a database built on a 1990s mainframe with a registry operating on a modern SQL system requires significant technical effort. Governments face considerable challenges when migrating from legacy systems, as these often contain critical data and perform essential functions (OECD Observatory of Public Sector Innovation, n.d.^[26]).

Artificial intelligence can help address some of these technical constraints by enabling automated data matching across systems, supporting the harmonisation of different classifications and facilitating the integration of otherwise incompatible datasets. However, these tools remain dependent on minimum levels of data quality and standardisation, and cannot compensate for fundamentally fragmented or poorly structured data environments.

Replacing legacy systems is expensive and carries operational risks. They often contain essential data, and any migration requires careful testing to avoid data loss or service interruptions. At the same time, keeping these systems in place creates long-term problems: older platforms keep data isolated, are difficult to update, and limit the level of interoperability needed for effective policy use. While AI-based tools can support aspects of system integration or documentation of legacy environments, they do not eliminate the need for structural modernisation and investment in interoperable data architectures. Nonetheless, periodic system renewal can also serve as an opportunity to re-examine administrative processes and the data collected, improving the timeliness and relevance of data collection, though the complexity of such programmes should not be underestimated.

5.4.2. Lack of common standards and definitions

A related challenge is the absence of common data standards and definitions. Education and training systems often rely on different identifiers for the same entities – for example, one database may use a national ID number for individuals, while another uses a student enrolment number, or they may classify fields of study in different ways. Inconsistent definitions create difficulties when datasets are merged, increasing the risk of errors or misinterpretation. For instance, if “completion” of a training programme is defined differently across providers, aggregate statistics on completion rates will lack reliability. A desirable way to address these issues is through the adoption of standardised classifications, such as ISCED for education levels or international occupation codes, which can strengthen data comparability and facilitate linkage across systems. Without such standards, even basic questions, such as how many people received training in a given year, can be difficult to answer consistently, as regions may count hours of instruction in some cases and numbers of participants in others. Some countries have also established sector-specific national data standards; Australia’s AVETMISS standard, for example, defines consistent reporting requirements for vocational education and training data across jurisdictions, supporting comparability and continuous improvements in data quality and timeliness.

Some countries have gone further by creating base registries – centralised databases that provide core information such as individuals, businesses or qualifications. Box 4 illustrates how Estonia addressed these challenges through a fully integrated education information system. The principle is that all government systems reference the same master data, ensuring consistency and avoiding duplication. At the same time, the existence of a single verified personal identifier increases the potential for re-identification if security is compromised, reinforcing the need for robust and continuously updated privacy safeguards. For example, a national population registry can serve as the single source of verified personal identifiers, which are then used across education, labour and social protection databases. Australia’s Unique Student Identifier (USI) plays a similar role for vocational and tertiary education, enabling training activity to be tracked across providers and over time. A concrete example of the use of standardised identifiers is presented also in, which outlines Chile’s approach to integrating education and labour data.

Box 4. Country examples: Estonia and Chile

Estonian Education Information System (Eesti hariduse infosüsteem, EHIS)

In 2004, Estonia implemented the EHIS, which has since served as an international reference. This database integrates key components of the education system, including schools, teachers, students, exams, qualifications, and curriculum materials. By connecting directly with schools, it ensures high data quality. Another major advantage is that it enables the longitudinal tracking of students over time. As a result, the data have become more granular, detailed, and regularly updated.

A further efficiency mechanism lies in the EHIS's connection to other regulated individual databases in areas such as health and employment. This linkage is made possible through the X-Tee – also known as X-Road – a “*technological and organisational environment enabling a secure Internet-based data exchange between information systems*”.

Chile's national identifiers (RUT and MRUN)

Chile has developed a robust system of unique identifiers that enables the integration of information across multiple policy domains. The Rol Único Tributario (RUT) is a unique number assigned to every citizen and resident, used primarily for tax and administrative purposes, but also widely adopted across the public and private sectors. The RUT functions as a single identifier that allows data from education, health, social protection and employment systems to be linked reliably to the same individual.

In addition, the education sector developed the Matrícula Rol Único Nacional (MRUN), a unique identifier for students enrolled in the education system. The MRUN is linked to the RUT, ensuring that education records can be connected with other administrative data sources. For example, student progression and completion data can be combined with labour market outcomes captured by social security or tax records. This linkage makes it possible to evaluate the effectiveness of education programmes and to monitor transitions from school to work with greater precision.

The use of RUT and MRUN illustrates how base registries or central reference databases can enhance consistency, reduce duplication, and enable more effective policy evaluation. By relying on a single verified identifier across institutions, Chile has increased its capacity to generate integrated evidence on skills development and outcomes. However, the reliance on a single identifier across systems also heightens long-term privacy risks, underscoring the importance of continuously strengthening safeguards.

Source: OECD (2020^[27]), *Strengthening the Governance of Skills Systems: Lessons from Six OECD Countries*, <https://doi.org/10.1787/3a4bb6ea-en>; Information System Authority (2019^[28]), *Data Exchange Layer X-tee*, www.ria.ee/en/state-information-system/data-exchange-platforms/data-exchange-layer-x-tee.

5.4.3. Limited use of APIs and real-time data exchange

Another important technical barrier is the limited availability of application programming interfaces (APIs) and real-time data access mechanisms in education and training systems. APIs are tools that allow different databases and applications to communicate with one another in a secure and automated way. In practice, this means that data held in one system – for example, student performance records – could be directly linked with information from another, such as local labour market indicators, without the need for repeated manual extraction and cleaning. When APIs are missing or underdeveloped, data integration becomes slow, labour-intensive and prone to errors, often requiring staff to download, reformat and merge files from multiple sources. This reduces the timeliness and reliability of the information available to policymakers.

More regular and automated flows of information would be particularly valuable for monitoring key indicators such as employability outcomes of VET graduates, early school leaving and dropout rates, enrolment in lifelong learning programmes, or the alignment of training provision with local labour market demand. Without such mechanisms, valuable data remains underused, and analytical capacity is diverted to data preparation rather than policy analysis. APIs themselves also require governance: mechanisms to validate use cases, define interface contracts and ensure that protected data are shared only under appropriate conditions. More broadly, interoperability should encompass data traceability and lifecycle management, so that when data are transformed or combined to produce indicators, their origin and the transformations applied remain auditable.

5.4.4. Cybersecurity and data resilience

Cybersecurity represents a critical challenge in the development of data-driven skills systems. As databases become increasingly interconnected, their exposure to cyber risks also grows (OECD, 2023^[14]; OECD, 2023^[29]). A breach of sensitive information – such as student or trainee records – could undermine public trust and significantly delay progress in data integration efforts. Concerns about such risks often make policymakers and IT departments cautious about linking systems, fearing that broader access may increase vulnerability. Insufficient protection against cyberattacks remains a pressing issue, with potential legal, operational and reputational consequences.

Data resilience poses a distinct set of challenges. Many older systems lack robust backup and disaster recovery mechanisms, creating a risk of prolonged downtime or even data loss in the event of system failure, whether caused by technical faults, natural disasters or human error. Building resilience – ensuring that data systems can withstand disruptions and recover rapidly – is essential for the continuity of evidence-based policymaking. If a key education or labour database were to become unavailable during a critical period, such as budget planning, it could directly compromise decision making. Addressing resilience requires sustained investment in infrastructure redundancy, regular testing of recovery procedures and clear protocols for maintaining analytical capacity during disruptions.

6 Conclusion

Building stronger and more integrated data systems for education, employment and skills is both necessary and challenging. The evidence presented in this paper shows that improved data can enable clearer and more equitable resource allocation, strengthen accountability and support more resilient and responsive skills systems. However, achieving these gains requires sustained investment and co-ordinated reform across four interdependent areas: institutional, governance and financing arrangements, human-capital and analytical capability, legal and regulatory frameworks, and technical and interoperability infrastructure. Progress in any one area alone is insufficient; each pillar must advance together to generate real value.

Improving data systems is not primarily a matter of collecting more information. In many countries, substantial amounts of relevant data already exist but remain underutilised because they are fragmented across agencies, constrained by restrictive rules, stored in incompatible systems, or inaccessible to decision makers. The priority is therefore to unlock and integrate existing data assets through better co-ordination, improved interoperability, strengthened analytical capacity and governance models that encourage evidence use rather than reinforce silos. Effective data systems depend as much on people and institutions as on technology.

Investing in these capabilities entails real financial and organisational costs. Modernising IT infrastructure, building secure data environments, establishing legal frameworks that enable safe data sharing, and developing analytical skills within the public sector require long-term commitments. These investments should therefore be supported by rigorous cost – benefit analysis. Many benefits, such as improved targeting, reduced duplication and early identification of ineffective programmes, accumulate over time rather than deliver immediate gains. Policymakers must balance short-term efficiencies with longer-term system transformation.

Digital technologies, including artificial intelligence, are rapidly expanding the potential to extract value from skills data. They can support more granular analysis, improve forecasting of skills needs, automate data processing and enhance the targeting of education and training investments. In this sense, AI can act as a multiplier of the returns to data systems, allowing governments to move from descriptive analysis to more predictive and adaptive policymaking. However, these opportunities also come with important risks, including biases embedded in algorithms, lack of transparency in decision making processes, and increased exposure to privacy and cybersecurity threats. AI outputs should therefore remain explainable, auditable and subject to human oversight, and correlations generated by these tools should not be presented as causal conclusions. Without strong data governance frameworks and high-quality underlying data, the use of AI may amplify existing weaknesses rather than address them.

Ultimately, the payoff from investing in stronger data and analytical systems is potentially large. Well-designed reforms can improve policy effectiveness, optimise public spending and strengthen labour market and social outcomes. The central challenge is to invest strategically, sequencing reforms to ensure alignment across the four pillars and building the institutional capacity required to translate data into better decisions.

Annex A. Types of data sources for decision making

Population and household surveys

Cross-sectional household and individual surveys

Cross-sectional household and individual surveys are widely used instruments for understanding the educational, social and economic characteristics of a population. These surveys collect information at a single point in time from nationally or regionally representative samples, providing a detailed snapshot of key elements such as access to education, levels of educational attainment, patterns of school attendance, reasons for dropout, household socio-economic characteristics, living conditions and demographic profiles. They constitute a foundational source for describing system performance and identifying disparities across groups and territories.

Examples of well-established cross-sectional household surveys include the Living Standards Measurement Study (LSMS) series of the World Bank, Demographic and Health Surveys (DHS) and UNICEF's Multiple Indicator Cluster Surveys (MICS). Many countries also operate national cross-sectional household surveys that integrate education modules, such as CASEN in Chile, ENIGH in Mexico, GEIH (household module) in Colombia, and Enquête Revenus Fiscaux et Sociaux in France. These instruments are frequently used to complement administrative education statistics and provide broader contextual information not available from available records.

Cross-sectional surveys are particularly valuable for describing access, participation and attainment in education, estimating socio-economic, gender and regional inequalities, and identifying population groups facing barriers to educational opportunities. They support analysis of the relationship between education and social outcomes such as income levels, poverty, health and early childhood development. These insights play an important role in informing priorities for public investment, designing targeting mechanisms and guiding social policy decisions.

However, from a decision making standpoint, it is important to recognise the analytical limitations of this type of data. Because they observe different individuals at a single moment rather than following the same people over time, cross-sectional surveys cannot provide evidence on educational or labour market trajectories or on how outcomes evolve in response to policy reforms. They are not suitable for measuring the effects of programmes or interventions, as they cannot distinguish policy influence from pre-existing differences between individuals, unobservable characteristics or broader macroeconomic conditions. As a result, analyses based solely on cross-sectional data can identify associations but cannot support credible causal conclusions about the impact of policies or programmes.

In practice, cross-sectional household and individual surveys offer significant value due to their representativeness, breadth of thematic coverage and periodic availability. They are particularly appropriate for diagnosing system-level inequalities, identifying priority groups and informing resource allocation. However, they are insufficient on their own for evaluating policy effectiveness, estimating programme impact or supporting high-stakes decisions such as scaling, redesigning or discontinuing

interventions. For purposes requiring causal inference or trajectory analysis, they must be complemented by longitudinal, administrative or experimental evidence.

Labour Force Surveys

Labour Force Surveys (LFS) are nationally representative surveys implemented at regular intervals – typically quarterly or monthly – to monitor labour market conditions and employment dynamics within the working-age population. They collect detailed information on employment status, unemployment, job search activity, occupation, industry sector, type of employment contract, hours worked and forms of labour market participation. Because of their frequency and consistency, LFS datasets are a primary reference for understanding labour market performance and for designing employment and workforce-related policies.

Examples include the European Union Labour Force Survey co-ordinated by Eurostat, the Current Population Survey (CPS) in the United States, the Labour Force Survey (LFS) in Canada, Enquête Emploi in France, ENOE in Mexico, GEIH (labour module) in Colombia, and similar instruments administered by national statistical offices across OECD and non-OECD countries. These surveys are widely used by labour ministries, public employment services and social security systems to track labour market dynamics, inform the design of employment and training programmes and identify groups requiring targeted support.

Labour Force Surveys are particularly valuable for understanding employment trajectories at the system level, examining unemployment patterns across demographic groups, analysing sectoral and regional labour demand, and identifying groups facing labour market barriers such as youth, women, migrants or older workers. They enable assessment of aggregate transitions between education and work, tracking indicators such as youth NEET rates (Not in Employment, Education or Training), inform the planning of training supply and support forecasting of workforce needs. Their ability to measure changes regularly makes LFS an important tool for policy response during economic shocks, recovery periods and structural labour market shifts.

However, there are important limitations to consider when using LFS data for policy decisions. Most Labour Force Surveys are cross-sectional, meaning they provide a snapshot of different individuals at a single point in time. Some operate as short rotating panels, but these typically follow respondents only for a limited number of waves and do not capture long-term changes. As a result, LFS datasets cannot observe individuals over extended periods or assess employment trajectories before and after a policy intervention, which limits their capacity to support analysis of long-term outcomes or causal effects.

Labour Force Surveys can describe correlations between qualifications and employment outcomes, but they cannot determine whether a particular programme, policy or reform caused changes in employment, productivity or earnings. Indicators captured in LFS are strongly influenced by broader macroeconomic fluctuations, demographic changes and institutional environments, which means that observed movements cannot be interpreted as policy effects without complementary evidence. In addition, LFS datasets generally lack granular information about programme characteristics, quality of training, employer behaviour or workplace practices, limiting their ability to explain why labour outcomes occur.

In practical terms, Labour Force Surveys are highly suitable for system-level labour analysis, labour market monitoring, workforce planning, identification of vulnerable groups and the design of employment and reskilling policies. They provide an essential evidence base for strategic decisions regarding labour supply, sectoral training priorities, and the organisation of public employment services. However, they are not sufficient for evaluating the effectiveness of specific education or employment programmes, nor for supporting high-stakes investment decisions, programme expansion or closure. For these purposes, LFS should be complemented by longitudinal data, administrative datasets, tracer studies or experimental and quasi-experimental evaluations.

Administrative and assessment data

Administrative data

Administrative education data are collected routinely by education systems for operational and regulatory purposes. These datasets are typically derived from school, college or training institution records and include detailed information on enrolment, attendance, progression, completion, staffing, infrastructure and financing. Many countries consolidate these data into Education Management Information Systems (EMIS), which centralise information on students, institutions and programmes to support planning and system monitoring. Examples include Estonia's EHIS, Norway's VIGO and National Education Database, and Chile's SIGE. Other countries have developed dedicated systems that support the connection, management and analysis of national education data. New Zealand's National Student Index is one such example. These systems typically integrate enrolment, attendance, progression and institutional data, allowing administrations to monitor participation, track pathways and allocate resources more effectively.

Administrative datasets are particularly valuable for monitoring participation and progression within the education and training system, analysing dropout, repetition, transitions between levels and completion rates, and understanding the allocation of financial and human resources across institutions and regions. These data support evidence-based planning, including forecasting demand for places in specific tracks or regions, adjusting funding formulas, and identifying institutions requiring targeted support. When integration across datasets is possible, administrative records can also enable analysis of medium- and long-term trajectories and provide the foundation for more robust evaluation designs.

However, administrative datasets also present important limitations that policymakers should consider. Because they are designed primarily for system management rather than analytical purposes, they often lack detailed information about students' socio-economic backgrounds, learning experiences or the quality of teaching and curriculum delivery. This limits the extent to which observed patterns can be interpreted or connected directly to policy actions. While administrative data enable strong descriptive analysis and trend monitoring, they do not provide causal evidence on the effects of reforms or programmes unless combined with appropriate evaluation designs or longitudinal linkages. Changes observed in administrative indicators may reflect multiple simultaneous influences – such as demographic shifts, economic conditions, classification changes or institutional incentives – rather than specific policy interventions.

Another consideration is data quality. As these datasets depend on reporting by institutions, accuracy may vary – incentive structures, workload pressures or inconsistent definitions can all introduce measurement discrepancies. Separately, the ability to link administrative education data with labour, tax or social protection systems depends on legal, technical and governance capacity, which may limit their potential for integrated analysis (see Section 5).

Standardised learning assessments and skills surveys

Standardised learning assessments and skills surveys are instruments designed to measure what learners and adults know and can do, using common frameworks, comparable scoring and consistent administration procedures. They enable meaningful comparisons across individuals, institutions, regions and countries, and support monitoring of performance over time. These assessments may focus on academic learning in school-aged populations or on foundational and workplace-relevant skills in adults.

Examples of large-scale international learning assessments include the OECD Programme for International Student Assessment (PISA), which evaluates 15-year-old students' proficiency in reading, mathematics and science; TIMSS and PIRLS, which assess mathematics, science and reading respectively in school contexts; and national assessment systems used for certification or progression, such as the baccalauréat in France, ENEM in Brazil, the Saber assessments administered by ICFES in Colombia, or high-stakes end-of-school examinations in many OECD and non-OECD countries. For adults,

the leading example is the OECD Programme for the International Assessment of Adult Competencies (PIAAC), which measures literacy, numeracy and problem-solving in technology-rich environments among individuals aged 16-65. PIAAC also includes detailed background questionnaires on education, employment, job tasks and training participation. Countries also field national skills surveys to fill specific evidence gaps; Australia's Foundation Skills Study, for instance, provides contemporary direct data on adult literacy and numeracy.

These assessments are particularly valuable for identifying disparities in learning and skill development across socio-economic groups, genders, regions and school types, and for analysing how performance changes between cohorts or over time. They provide evidence for system-level policy prioritisation, support curriculum reform discussions, and offer comparative benchmarks that help countries understand their performance relative to peers. Skills surveys such as PIAAC also allow analysis of the relationship between skill levels and labour market outcomes, informing adult learning, reskilling and workforce development strategies.

However, it is essential to recognise the limitations of these tools when used to inform policy decisions. Standardised assessments focus primarily on the skills that can be measured reliably and comparably at scale. While some newer instruments include elements that capture broader competencies – such as socio-emotional skills, collaborative problem-solving or aspects of creativity – the scope remains necessarily constrained. Interest in measuring socio-emotional skills systematically is nonetheless growing, given their demonstrated links to educational retention, employability and well-being (OECD, 2025^[10]). Many complex or context-dependent competencies, including professional practice or teamwork in real environments, are difficult to assess through standardised large-scale testing.

While standardised learning assessments and skills surveys support analysis of associations between contextual factors and outcomes, they are not designed to determine whether a specific policy or programme caused the observed results. Many reforms operate simultaneously, and test score changes may reflect demographic shifts, implementation variation or external economic conditions rather than the direct effect of an intervention. Because most assessments are periodic rather than continuous, they are not suitable for short-term monitoring or rapid adjustment strategies.

Additionally, these instruments provide limited insight into how learning processes occur or why certain policies succeed or fail, since they do not collect detailed information on programme quality, pedagogical practices, resource allocation or institutional context. For these reasons, assessment results must be interpreted carefully, and policy conclusions should be supported with complementary evidence from administrative data, longitudinal datasets or rigorous evaluation methods.

Programme-focussed data

Longitudinal and panel data

Longitudinal and panel data follow the same individuals, households or institutions over an extended period of time, collecting information at multiple points. Unlike cross-sectional surveys, which provide a single snapshot, longitudinal datasets enable analysis of trajectories, transitions and the evolution of outcomes before, during and after policies or interventions. These datasets may be formed through repeated surveys, linked administrative records, cohort studies or labour market panels, and can cover domains such as education, employment, income, social protection and health.

Prominent examples include the 1970 British Cohort Study (BCS70), which has followed over 17 000 individuals born in a single week of 1970 in England, Scotland and Wales. Initially focussed on perinatal health, it has evolved to provide rich evidence on health, education, labour market participation and socio-economic outcomes across the life course. The British Household Panel Survey (BHPS) tracked approximately 10 000 households annually from 1991 to 2009 before being incorporated into

Understanding Society, one of the world's largest household longitudinal studies. Other examples include the German Socio-Economic Panel (SOEP), the US National Longitudinal Surveys (NLS) and registry-based longitudinal data systems such as Denmark's DREAM database.

Longitudinal and panel data are particularly valuable for understanding how individuals progress through education and employment, how early disadvantages compound over time, and how participation in programmes is associated with medium- and long-term results. They support analysis of transitions – such as from school to work or from unemployment to stable employment – and allow examination of persistence, timing and turning points rather than static differences. These datasets provide a stronger basis for policy evaluation because they enable comparison of individuals before and after interventions and facilitate controlling for time-invariant characteristics.

However, longitudinal studies also involve important limitations and trade-offs. They typically have a narrower or more specific scope than survey-based national systems, as they often focus on particular cohorts (for example, youth, birth cohorts or specific programme participants). Their operational and follow-up costs are high, requiring long-term financial commitment, technical capacity, ethical frameworks and data governance. Attrition over time can reduce representativeness, particularly affecting disadvantaged groups, and fragmentation across systems may limit opportunities for interoperability and linkage. Importantly, although longitudinal data strengthen analytical potential, they do not generate causal evidence on their own. Credible causal attribution still requires a clearly defined evaluation design, suitable comparison groups and methods that can isolate policy effects from other factors. An additional consideration is timeliness: the long follow-up periods required to capture meaningful trajectories mean that by the time robust findings emerge, the skills needs of the economy may have evolved, potentially limiting the direct policy relevance of the results.

Tracer studies of graduates and programme participants

Tracer studies are surveys or follow-up studies conducted with individuals after they complete an educational programme, training intervention or social or employment support initiative. They typically collect information several months or years after exit, with the objective of understanding participants' experiences, labour market integration, perceptions of programme quality and relevance, and short- to medium-term outcomes such as employment, earnings, job stability or further study. Tracer studies are commonly employed in vocational education and training (VET), higher education, active labour market programmes and social protection initiatives, particularly when institutions seek to understand participants' early outcomes and experiences after completion.

Tracer studies are valuable because they provide programme-specific evidence that is not available from national Labour Force Surveys or administrative systems. For example, they generate detailed information on participants' perceptions of training quality, relevance of curricula to job requirements, and the extent to which skills acquired are used in the workplace. They support understanding of early labour market transitions, including entry pathways, employment barriers, and job quality. For providers, tracer studies inform programme improvement, career guidance services and employer engagement strategies.

However, important limitations must be recognised when using tracer study evidence for policy decisions. Tracer studies generally lack comparison groups, making it difficult to distinguish the effect of the programme from participants' prior characteristics or labour market conditions. Without a counterfactual, observed employment or earnings outcomes cannot be interpreted as causal effects. Response rates may be uneven, and results may be biased if individuals with weaker outcomes opt out of follow-up, or if respondents differ systematically from non-respondents. Tracer studies also have a restricted scope, typically covering graduates of a single programme, institution or intervention rather than providing a system-wide view. Furthermore, they are costly and operationally demanding, as they require tracking participants over time, maintaining contact, ensuring data protection compliance and sometimes using mixed-mode follow-up strategies.

Evaluative and qualitative methods

Employer and enterprise surveys

Employer and enterprise surveys collect information directly from firms and industry representatives regarding labour demand, recruitment challenges, skills needs, training practices and the adoption of new technologies. They typically target business owners or human resource managers and may operate at national, regional or sectoral level. Because they capture employer perspectives, they offer insights into labour market requirements that cannot be observed through household or education data.

These surveys are especially valuable for identifying emerging skills shortages, estimating recruitment difficulties, and understanding how employers perceive the preparedness of graduates and workers. They help policymakers to align training supply with labour market demand, prioritise occupations with strong job prospects and adjust qualification frameworks or curricula in vocational and higher education. They can also provide information on firms' own investment in training and internal workforce development.

However, several limitations are important to recognise when using employer survey evidence in policymaking. First, many employer surveys are sectoral, meaning that results are often not representative of the economy as a whole, and generalisation beyond the surveyed sector can lead to misleading conclusions. Second, methodologies, definitions and timing vary considerably across surveys and countries, which means results are rarely comparable across sectors or internationally. While international comparability may be less critical for a policymaker focussed on a specific domestic sector, inconsistent definitions can still limit the ability to benchmark performance or identify transferable practices. Third, responses often reflect employers' perceptions, which may not coincide with objective skill requirements or performance indicators, and may be influenced by short-term pressures. Large firms tend to be over-represented, while micro-enterprises and informal labour markets are frequently under-covered. Finally, employer surveys are generally not designed to establish whether specific programmes or reforms caused observed skills improvements, though they can capture employers' perceptions of programme effectiveness, which offer a complementary – if subjective – evidence base.

Evaluative and qualitative methods

Data from experimental and pilot programmes

Experimental and pilot programme data are generated through interventions that are intentionally designed to test new policies, programmes or delivery models in a controlled or semi-controlled environment. These initiatives often include Randomised Controlled Trials (RCTs) or quasi-experimental designs, in which participants are assigned to treatment and comparison groups. This allows the resulting data to support credible conclusions about cause – effect relationships.

Well-known international examples include the Jamaica Early Childhood Stimulation Study, one of the most influential RCTs in social policy. Beginning in the 1980s, it followed disadvantaged children who received early stimulation and parental coaching, comparing them with a control group. The study generated high-quality longitudinal data and demonstrated large and persistent gains in cognitive outcomes, schooling attainment and adult earnings, decades after the intervention (Gertler et al., 2021^[30]; Gertler et al., 2014^[31]). It remains a reference case worldwide for evidence-based investment decisions in early childhood development. It is worth noting, however, that the intervention targeted stunted children in a disadvantaged community and included a nutritional component, which may account for a significant part of the observed gains. Caution is therefore warranted when generalising these results to broader early childhood programmes, particularly in high-income settings where stunting is less prevalent. Other

examples include experimental evaluations of conditional cash transfer programmes and active labour market policies in Latin America (Escudero et al., 2017^[32]).

These initiatives typically generate datasets that include detailed baseline characteristics, implementation records, and outcome measures such as learning gains, employment rates, job stability or earnings, often with follow-up data at multiple time points. The primary advantage of this type of data is its high internal validity, which allows policymakers to interpret observed differences as effects of the intervention rather than consequences of external factors. These datasets are also well suited for cost – effectiveness and cost – benefit analyses, supporting transparent and accountable resource allocation.

However, experimental and pilot programme data also have important limitations (Deaton and Cartwright, 2018^[33]). Because these initiatives are usually implemented in specific locations, target groups or institutional contexts, the results often have limited external validity and may not generalise to national systems or broader populations. Sample sizes are typically smaller than those in national datasets, and follow-up periods may be short, which restricts the ability to capture long-term outcomes or spill-over effects. In addition, many RCTs and pilot interventions take place under controlled or highly supported conditions, with higher implementation quality, closer monitoring and stronger delivery standards than those that may be feasible at scale. As a result, effects observed in pilot settings may not fully reflect the outcomes that would occur once the intervention is expanded to more diverse institutions or delivered under routine system conditions. These studies are also costly, require advanced methodological and data management capacity, and depend on strong partnerships between institutions.

Qualitative data and case study research

Qualitative data and case study research involve systematic collection of non-numerical information through methods such as interviews, focus groups, field observations, classroom or workplace observations, and document analysis. Rather than aiming for statistical representativeness, qualitative approaches seek to understand mechanisms, behaviours, perceptions and implementation processes that are not visible in quantitative datasets. They provide depth and contextual understanding that support interpretation of quantitative results and inform policy and programme design.

Qualitative evidence is particularly valuable for exploring how and why policies and programmes operate as they do in real contexts. It can reveal perspectives from groups that may be under-represented or invisible in administrative datasets or survey samples, such as informal workers, learners outside the formal system or individuals with complex support needs. It can illuminate implementation barriers, institutional incentives, professional cultures and unintended consequences, providing insight into factors that influence programme success or failure. It also helps explain divergent results in quantitative analysis and generate hypotheses for further study.

However, qualitative evidence has natural limitations. Because it is not statistically representative, results cannot be generalised to the broader population without caution. Findings depend heavily on sampling strategies, researchers' interpretation and participants' willingness to share information. Qualitative approaches are not designed to measure programme impact or estimate effect sizes in the statistical sense. However, consistent patterns in participant-reported experiences can offer complementary evidence on mechanisms and perceived outcomes. They should not be used alone as a basis for high-stakes investment or reform decisions, but can strengthen the interpretation of quantitative findings. Collection and analysis can also be resource-intensive when implemented at scale.

Emerging data sources

Big data, learning analytics and online job postings

Big data and learning analytics refer to large, high-frequency datasets generated through digital platforms and administrative technologies. These include data produced by online learning and training platforms, mobile applications and professional networking platforms. Because they capture real-time behavioural information rather than self-reported responses, they can provide highly detailed insight into participation patterns, engagement and performance.

A particularly important and rapidly growing source within this category is online job postings data, generated from digital job advertising platforms and aggregated labour market intelligence systems. These datasets collect information on vacancies, occupational profiles, required skills, credentials, technologies used and wage offers advertised by employers. OECD examples of research using these data, including analyses based on online job postings, include recent work by (Kuczera, 2025^[34]; OECD, 2025^[35]).

Big data, learning analytics and job postings are especially valuable for detecting emerging labour market trends, identifying skills in growing demand and informing curriculum modernisation and training priorities. Online postings provide near real-time insight into technology adoption, changing occupational profiles and local or sectoral shortages. Similarly, education platforms can track learner engagement, progression, completion and interaction with content, enabling adaptive learning models, early warning systems and targeted support.

However, these data have important limitations for policymaking. Online job postings capture only demand that is visible in digital markets, which means they overrepresent formal, large and high-skilled employers while underrepresenting micro-enterprises, informal sectors and rural labour markets. Not all postings correspond to actual hiring intentions, and duplicated postings or automated listings can distort volume and trend interpretation. In addition, postings reflect employers' expectations and preferences, which do not necessarily match actual job requirements or workplace practices. Learning analytics datasets contain only the behaviour of platform users and therefore exclude individuals without access or digital engagement, generating selection bias that can mask inequality. These datasets do not measure outcomes such as earnings, productivity or long-term employment and they cannot establish causality.

For these reasons, big data should be treated as a strictly complementary source: useful for horizon scanning and timely signals, but not a substitute for representative statistics or administrative records, and always with their limitations clearly documented.

References

- Burtscher, M., S. Piano and B. Welby (2024), “Developing skills for digital government: A review of good practices across OECD governments”, *OECD Social, Employment and Migration Working Papers*, No. 303, OECD Publishing, Paris, <https://doi.org/10.1787/f4dab2e9-en>. [18]
- Deaton, A. and N. Cartwright (2018), “Understanding and misunderstanding randomized controlled trials”, *Social Science & Medicine*, Vol. 210, pp. 2-21, <https://doi.org/10.1016/j.socscimed.2017.12.005>. [33]
- Escudero, V. et al. (2017), “Active Labour Market Programmes in Latin America and the Caribbean: Evidence from a Meta Analysis”, *IZA – Institute of Labor Economics*, Vol. 11039, <https://docs.iza.org/dp11039.pdf>. [32]
- Gertler, P. et al. (2021), *Effect of the Jamaica Early Childhood Stimulation Intervention on Labor Market Outcomes at Age 31*, National Bureau of Economic Research, Cambridge, MA, <https://doi.org/10.3386/w29292>. [30]
- Gertler, P. et al. (2014), “Labor market returns to an early childhood stimulation intervention in Jamaica”, *Science*, Vol. 344/6187, pp. 998-1001, <https://doi.org/10.1126/science.1251178>. [31]
- Government of Canada (2023), *CSPS Digital Academy*, <https://doi.org/10.1787/f4dab2e9-en> (accessed on 15 September 2025). [19]
- GovTech Lab (2024), *What is GovTech Lab?*, <https://govtechlab.lt/about/> (accessed on 15 May 2026). [20]
- Information System Authority (2019), *Data Exchange Layer X-tee*, <https://www.ria.ee/en/state-information-system/data-exchange-platforms/data-exchange-layer-x-tee>. [28]
- Kuczera, M. (2025), “Vocational education and training (VET) and the green transition: Insights from labour market data”, *OECD Social, Employment and Migration Working Papers*, No. 327, OECD Publishing, Paris, <https://doi.org/10.1787/02b7fcb1-en>. [34]
- Leseman, P. and P. Slot (2025), “Strong early childhood education and care systems for the future: A conceptual framework for thematic analyses”, *OECD Education Working Papers*, No. 329, OECD Publishing, Paris, <https://doi.org/10.1787/67d0a97f-en>. [5]
- Ministry of Personnel Management (2023), *Talent Development Platform*, <https://www.learning.go.kr/intro/board/viewPlatformIntroduce.do> (accessed on 15 September 2025). [17]

- OECD (2025), *Education at a Glance 2025: OECD Indicators*, OECD Publishing, Paris, <https://doi.org/10.1787/1c0d9c79-en>. [8]
- OECD (2025), *Empowering the Workforce in the Context of a Skills-First Approach*, OECD Skills Studies, OECD Publishing, Paris, <https://doi.org/10.1787/345b6528-en>. [35]
- OECD (2025), *Skills that Matter for Success and Well-being in Adulthood: Evidence on Adults' Social and Emotional Skills from the 2023 Survey of Adult Skills*, OECD Skills Studies, OECD Publishing, Paris, <https://doi.org/10.1787/6e318286-en>. [10]
- OECD (2025), *Trends in Adult Learning: New Data from the 2023 Survey of Adult Skills*, Getting Skills Right, OECD Publishing, Paris, <https://doi.org/10.1787/ec0624a6-en>. [7]
- OECD (2024), *Bridging Talent Shortages in Tech: Skills-first Hiring, Micro-credentials and Inclusive Outreach*, Getting Skills Right, OECD Publishing, Paris, <https://doi.org/10.1787/f35da44f-en>. [15]
- OECD (2024), *Enabling Digital Innovation in Government: The OECD GovTech Policy Framework*, OECD Digital Government Studies, OECD Publishing, Paris, <https://doi.org/10.1787/a51eb9b2-en>. [21]
- OECD (2024), *Insights from Skills Strategies in the European Union: Lessons Learnt for Developing and Implementing Effective Skills Policies*, OECD Skills Studies, OECD Publishing, Paris, <https://doi.org/10.1787/0bf9e78e-en>. [11]
- OECD (2024), *Quality Matters: Strengthening the Quality Assurance of Adult Education and Training*, OECD Skills Studies, OECD Publishing, Paris, <https://doi.org/10.1787/f44a185b-en>. [3]
- OECD (2023), *Building a Skilled Cyber Security Workforce in Five Countries: Insights from Australia, Canada, New Zealand, United Kingdom, and United States*, OECD Skills Studies, OECD Publishing, Paris, <https://doi.org/10.1787/5fd44e6c-en>. [14]
- OECD (2023), *OECD Digital Education Outlook 2023: Towards an Effective Digital Education Ecosystem*, OECD Publishing, Paris, <https://doi.org/10.1787/c74f03de-en>. [29]
- OECD (2021), *Education at a Glance 2021: OECD Indicators*, OECD Publishing, Paris, <https://doi.org/10.1787/b35a14e5-en>. [9]
- OECD (2020), *Strengthening the Governance of Skills Systems: Lessons from Six OECD Countries*, OECD Skills Studies, OECD Publishing, Paris, <https://doi.org/10.1787/3a4bb6ea-en>. [27]
- OECD (2019), *Enhancing Access to and Sharing of Data: Reconciling Risks and Benefits for Data Re-use across Societies*, OECD Publishing, Paris, <https://doi.org/10.1787/276aaca8-en>. [24]
- OECD (2019), *OECD Skills Strategy 2019: Skills to Shape a Better Future*, OECD Publishing, Paris, <https://doi.org/10.1787/9789264313835-en>. [4]
- OECD (2019), *Providing Quality Early Childhood Education and Care: Results from the Starting Strong Survey 2018*, TALIS, OECD Publishing, Paris, <https://doi.org/10.1787/301005d1-en>. [6]
- OECD (2019), *The Path to Becoming a Data-Driven Public Sector*, OECD Digital Government Studies, OECD Publishing, Paris, <https://doi.org/10.1787/059814a7-en>. [1]

- OECD Observatory of Public Sector Innovation (n.d.), *Digital Transformation*, <https://oecd-opsi.org/guide/digital-transformation/> (accessed on 18 May 2026). [26]
- OECD Observatory of Public Sector Innovation (n.d.), *OECD Observatory of Public Sector Innovation (OPSI)*, <http://oecd-opsi.org> (accessed on 18 May 2026). [16]
- Reznikova, L., R. Labanino and D. McKee Mathews (2024), “Pooling our strengths: The power of stakeholder engagement in education and skills policy”, *OECD Social, Employment and Migration Working Papers*, No. 316, OECD Publishing, Paris, <https://doi.org/10.1787/b73697cb-en>. [2]
- Solita (2026), *SECLE supports the employment market with data*, <https://www.solita.fi/work/secle-supports-the-employment-market-with-data/> (accessed on 15 May 2026). [13]
- Stats NZ Tatauranga Aotearoa (2022), *Integrated Data Infrastructure*, <https://www.stats.govt.nz/integrated-data/integrated-data-infrastructure/> (accessed on 18 May 2026). [25]
- The Service Centre for Continuous Learning and Employment (2026), *About Us*, <https://www.jotpa.fi/en/about-us> (accessed on 15 May 2026). [12]
- UNESCO (2023), *Global Education Monitoring Report 2023: Technology in education: A tool on whose terms?*, UNESCO, <https://doi.org/10.54676/uzqv8501>. [23]
- van Ooijen, C., B. Ubaldi and B. Welby (2019), “A data-driven public sector: Enabling the strategic use of data for productive, inclusive and trustworthy governance”, *OECD Working Papers on Public Governance*, No. 33, OECD Publishing, Paris, <https://doi.org/10.1787/09ab162c-en>. [22]

Better skills data for smarter financing of education and training

Robust and integrated skills data are essential for effective policymaking, particularly in improving the allocation of public resources across education and training. Yet many governments face significant information gaps. Fragmented or underdeveloped data systems constrain the ability to assess programme effectiveness, prioritise spending, and target investments towards high-impact interventions. This paper examines how stronger skills data can support more evidence-based financing decisions. It introduces a framework grouping the main obstacles into four categories: institutional, governance and financing barriers; human-capital and analytical-capacity gaps; legal and regulatory constraints; and technical and interoperability challenges. By analysing these barriers and highlighting promising country practices, the paper provides a basis for identifying priorities and guiding reforms to strengthen skills data systems and improve investment in skills.